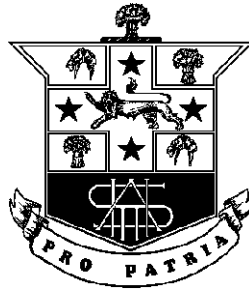


HURLSTONE AGRICULTURAL HIGH SCHOOL



YEAR 12

MATHEMATICS EXTENSION 2

2003

HSC COURSE

HALF YEARLY ~ ASSESSMENT TASK 2

EXAMINERS ~ J. DILLON, P. BICZO, S. GEE

GENERAL INSTRUCTIONS

- | | |
|---|--|
| <ul style="list-style-type: none">• READING TIME – 5 MINUTES.• WORKING TIME – TWO HOURS.• ATTEMPT ALL QUESTIONS.• QUESTIONS ARE OF EQUAL VALUE.• ALL NECESSARY WORKING SHOULD BE SHOWN IN EVERY QUESTION.• THIS PAPER CONTAINS SIX (6) QUESTIONS.• TOTAL MARKS – 90 MARKS | <ul style="list-style-type: none">• MARKS MAY NOT BE AWARDED FOR CARELESS OR BADLY ARRANGED WORK.• BOARD APPROVED CALCULATORS MAY BE USED.• EACH QUESTION IS TO BE STARTED ON A NEW PIECE OF PAPER.• THIS ASSESSMENT TASK MUST NOT BE REMOVED FROM THE EXAM CENTRE. |
|---|--|

STUDENT NAME: _____

TEACHER: _____

QUESTION 1:**MARKS**

- (a) If $A = 3 + 4i$ and $B = 2 - i$ express the following in the form $x + iy$ where $x, y \in \mathbb{R}$ $x, y \in \square$.
- (i) AB 1
- (ii) $\sqrt{A}$ 2
- (iii) $\frac{A}{B}$ 2
- (b) If $z = \sqrt{3} + i$
- (i) Find the exact values of $|z|$ and $\arg(z)$ 2
- (ii) By using your answers to (i) and De Moivre's Theorem, write z^5 in the form $x + iy$ where $x, y \in \mathbb{R}$. 2
- (c) On an Argand diagram, shade the region containing all the points representing the complex numbers z such that: 2
- $$1 \leq |z - 1| \leq 2 \text{ and } \frac{\pi}{4} < \arg(z - 1) < \frac{\pi}{2}$$
- (d) Explain algebraically, or geometrically, why the locus described by $\arg\left(\frac{z}{z-1}\right) = \frac{\pi}{2}$ is a circle. 2
- (e) Given that z and w represent two complex numbers, explain why $|z| + |w| \geq |z - w|$. 2

QUESTION 2: (START A NEW PIECE OF PAPER)

MARKS

- a) The equation $x^3 - 5x + 7 = 0$ has roots α, β, γ . 2

Find $\alpha^3 + \beta^3 + \gamma^3$.

- b) If $ax^3 + bx^2 + d = 0$ has a double root, show that $27a^2d + 4b^3 = 0$. 2

- c) Solve $2x^4 + 9x^3 + 6x^2 - 20x - 24 = 0$, if it has a root of multiplicity 3. 3

- d) The equation $x^4 + 4x^3 - 3x^2 - 4x + 2 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

Find the equations with roots

- i) $2\alpha, 2\beta, 2\gamma, 2\delta$. 2

- ii) $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}, \frac{1}{\delta^2}$. 2

- e) Given that $\cos 5\theta = 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta$

- (i) Solve the equation $16x^5 - 20x^3 + 5x - 1 = 0$ 2

- (ii) Hence, or otherwise, show that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = \frac{1}{2}$ 2

QUESTION 3: (START A NEW PIECE OF PAPER)

MARKS

- (a) A line through $P(a \sec \theta, b \tan \theta)$, perpendicular to the x - axis, meets an asymptote of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at Q . The normal at P meets the x - axis at G . 4

Show that GQ is at right angles to the asymptote.

(You may assume that the equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $P(a \sec \theta, b \tan \theta)$ is $ax \tan \theta + by \sec \theta = (a^2 + b^2) \sec \theta \tan \theta$.)

- (b) Consider the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a > b > 0$. 5

R and R' are the feet of the perpendiculars from the foci S and S' on to the tangent at $P(a \cos \theta, b \sin \theta)$.

Show that $SR \times S'R' = b^2$.

(You may assume the tangent to the ellipse at the point $P(a \cos \theta, b \sin \theta)$ has equation $bx \cos \theta + ay \sin \theta - ab = 0$.)

- (c) T is a variable point $(ct, \frac{c}{t})$ on the hyperbola $xy = c^2$. The perpendicular from the centre O to the tangent at T meets it in N .

(i) Find the coordinates of N . 1

(ii) Show that N lies on the curve $(x^2 + y^2)^2 = 4c^2 xy$. 2

- (d) Consider the curve $xy(x + y) + 16 = 0$. 3
Find the equation of the tangent to this curve at the point $P(-2, -2)$.

QUESTION 4: (START A NEW PIECE OF PAPER)

MARKS

- (a) Consider the function $f(x) = \frac{1-x}{x}$. On separate diagrams, sketch the graphs of the following functions / relations.

For each graph label any asymptotes.

(i) $y = f(x)$ 2

(ii) $y^2 = f(x)$ 3

- (b) Let $f(x) = -x^2 + 6x - 8$.

On separate diagrams, and without using calculus, sketch the following graphs.

Indicate clearly any asymptotes and intercepts with the axes.

(i) $y = f(x)$ 2

(ii) $y = |f(x)|$ 2

(iii) $y^2 = f(x)$ 2

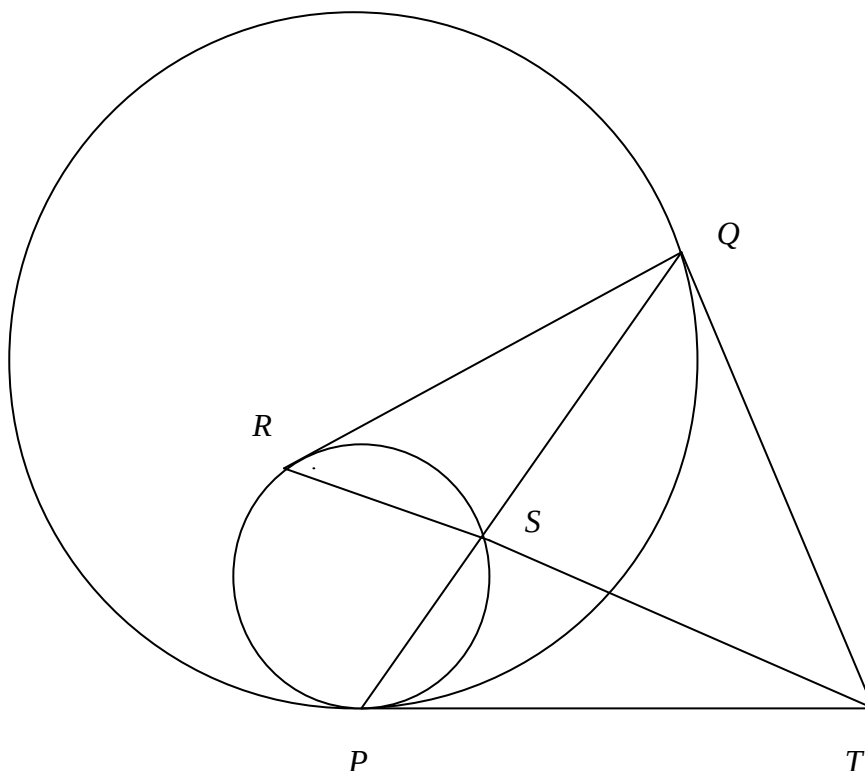
(iv) $y = \frac{1}{f(x)}$ 2

(v) $y = e^{f(x)}$ 2

QUESTION 5: (START A NEW PIECE OF PAPER)

MARKS

(a)



PT is the common tangent to the two circles at P .

QT is a tangent to the large circle at Q .

RQ is a tangent to the small circle at R .

RST and PSQ are straight.

- | | | |
|------|------------------------------------|---|
| (i) | Show that $PRQT$ is cyclic. | 3 |
| (ii) | Deduce RS bisects $\angle PRQ$. | 3 |
-
- | | | | |
|-----|------|---|---|
| (b) | (i) | By rationalising the numerator, prove that $\sqrt{n+1} - \sqrt{n} > \frac{1}{2\sqrt{n+1}}$. | 2 |
| | (ii) | Hence prove by induction that $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} < \sqrt{n}$, for $n \geq 7$. | 3 |
-
- | | | | |
|-----|-------|---|---|
| (c) | (i) | Find any stationary points on the curve $y = e^x - x - 1$ and determine their nature. | 2 |
| | (ii) | Sketch the curve. | 1 |
| | (iii) | Use the graph to show that $e^x \geq x + 1$ | 1 |

QUESTION 6: (START A NEW PIECE OF PAPER)**MARKS**

- (a) Given that the process of differentiation from first principles can be performed using

3

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

and Newton's expansion for e^x is

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

where $n! = n(n-1)(n-2)(n-3)\dots 3.2.1$,

show that $\frac{d}{dx}(e^x) = e^x$.

- (b) (i) Sketch $y = \frac{1}{x+1}$ for $x > -1$, showing where it cuts the y-axis and its asymptotes.

1

- (ii) Show that the area under the curve from $x = 0$ to $x = h$ is given by $\log_e(1+h)$.

1

- (iii) Show, with reasons, that $\frac{h}{1+h} < \log_e(1+h) < h$.

2

- (iv) Prove that $\lim_{h \rightarrow 0} \frac{\log_e(1+h)}{h} = 1$.

2

- (v) Using part (d) for $x > 0$, show $\lim_{x \rightarrow \infty} x \log_e\left(1 + \frac{1}{x}\right) = 1$

2

- (c) (i) Given that $a > 0$ and $b > 0$, prove that $\frac{a+b}{2} \geq \sqrt{ab}$

2

- (ii) Hence, or otherwise, given $a > 0$, $b > 0$, $c > 0$, $d > 0$ prove $\frac{a+b+c+d}{4} \geq \sqrt[4]{abcd}$

2

Question 1

$$(a) (i) AB = (3+4i)(2-i) \\ = 6-3i+8i-4i^2 \\ = 10+5i$$

$$(iii) \frac{A}{B} = \frac{3+4i}{2-i} \times \frac{2+i}{2+i} \\ = \frac{2+11i}{5} \\ = \frac{2}{5} + \frac{11}{5}i$$

$$(ii) \sqrt{A} = \sqrt{3+4i} = x+iy \\ 3+4i = x^2-y^2+2xyi \\ \therefore x^2-y^2 = 3 \quad \dots (1)$$

$$xy = 2 \quad \dots (2)$$

$$(2) \Rightarrow y = \frac{2}{x} \quad \dots (3)$$

$$(3) \rightarrow (1) \quad x^2 - \frac{4}{x^2} = 3$$

$$x^4 - 4 = 3x^2$$

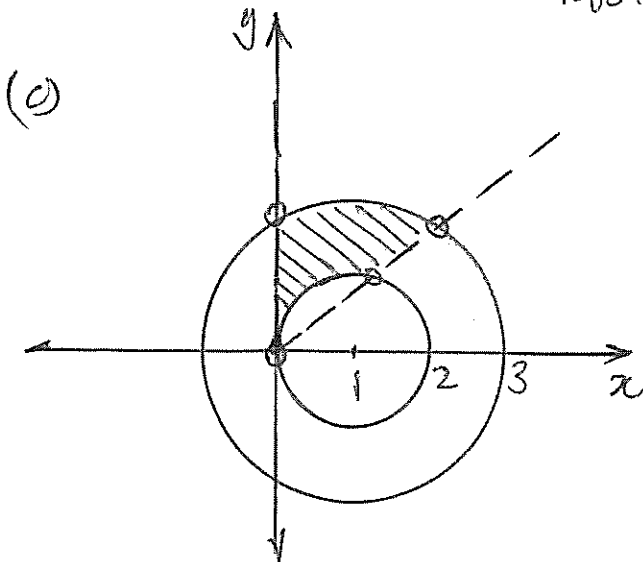
$$\therefore (x^2-4)(x^2+1) = 0$$

$$\therefore x = \pm 2, y = \pm 1$$

$$\therefore \sqrt{A} = \pm (2+i)$$

$$(b) (i) |z| = 2 \text{ and } \arg(z) = \frac{\pi}{6}$$

$$(ii) z^5 = \left(2 \operatorname{cis} \frac{\pi}{6}\right)^5 = 2^5 \operatorname{cis} \frac{5\pi}{6} \\ = 32 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) \\ = 32 \left(-\frac{\sqrt{3}}{2} + i \cdot \frac{1}{2} \right) \\ = -16\sqrt{3} + 16i$$



$$(d) \arg\left(\frac{z}{z-1}\right) = \frac{\pi}{2}$$

$$\arg z - \arg(z-1) = \frac{\pi}{2}$$

$$\tan^{-1} \frac{y}{x} - \tan^{-1} \frac{y}{x-1} = \frac{\pi}{2}$$

$$\frac{\frac{y}{x} - \frac{y}{x-1}}{1 + \frac{y}{x} \cdot \frac{y}{x-1}} = \tan \frac{\pi}{2}$$

$$\therefore 1 + \frac{y}{x} \cdot \frac{y}{x-1} = 0$$

$$x(x-1) + y^2 = 0$$

$$x^2 - x + y^2 = 0$$

Circle, centre $\left(\frac{1}{2}, 0\right)$ $\rightarrow \left(x - \frac{1}{2}\right)^2 + y^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$
radius = $\frac{1}{2}$

$$(e) \text{ Consider } (|z| + |w|)^2$$

$$= |z|^2 + |w|^2 + 2|z||w|$$

From the cosine rule

$$|z-w|^2 = |z|^2 + |w|^2 - 2|z||w|\cos\theta$$

$$\leq |z|^2 + |w|^2$$

$$\therefore |z|^2 + |w|^2 - 2|z||w|\cos\theta \leq |z|^2 + |w|^2 + 2|z||w|$$

$$\therefore |z-w|^2 \leq |z|^2 + |w|^2 + 2|z||w|$$

$$\therefore |z-w| \leq |z| + |w|$$

Question 2

$$(a) x^3 - 5x + 7 = 0$$

$$\alpha^3 - 5\alpha + 7 = 0$$

$$\beta^3 - 5\beta + 7 = 0$$

$$\gamma^3 - 5\gamma + 7 = 0$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 - 5(\alpha + \beta + \gamma) + 21 = 0$$

$$\alpha^3 + \beta^3 + \gamma^3 - 5(0) + 21 = 0$$

$$\alpha^3 + \beta^3 + \gamma^3 = -21$$

$$(c) P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24$$

$$P'(x) = 8x^3 + 27x^2 + 12x - 20$$

$$P''(x) = 24x^2 + 54x + 12$$

$$P''(x) = 0 \quad 24x^2 + 54x + 12 = 0$$

$$4x^2 + 9x + 2 = 0$$

$$(4x+1)(x+2) = 0$$

$$x = -2 \text{ or } -\frac{1}{4}$$

$$P'(-2) = 8(-2)^3 + 27(-2)^2 + 12(-2) - 20 = 0$$

$$P'(-\frac{1}{4}) = 8(-\frac{1}{4})^3 + 27(-\frac{1}{4})^2 + 12(-\frac{1}{4}) - 20 = -\frac{343}{16}$$

$$\# P(-2) = 2(-2)^4 + 9(-2)^3 + 6(-2)^2 - 20(-2) - 24 = 0$$

$$\therefore \text{Triple root is } x = -2$$

$$(b) P(x) = ax^3 + bx^2 + d$$

$$P'(x) = 3ax^2 + 2bx$$

$$= x(3ax + 2b)$$

$$P'(x) = 0 \Rightarrow x = 0 \text{ or } x = -\frac{2b}{3a}$$

NB $x = 0$ is not a solution to $P(x) = 0$

$$\therefore P(-\frac{2b}{3a}) = 0$$

$$\therefore a\left(-\frac{2b}{3a}\right)^3 + b\left(-\frac{2b}{3a}\right)^2 + d = 0$$

$$-\frac{8b^3}{27a^2} + \frac{4b^3}{9a^2} + d = 0$$

$$-8b^3 + 12b^3 + 27a^2d = 0$$

$$\therefore 27a^2d + 4b^3 = 0$$

Roots are $-2, -2, -2, \alpha$

$$\# \alpha - 6 = -4$$

$$\alpha = 2$$

$\therefore$ Roots are

$$x = -2, -2, -2, 2.$$

(d) (i) let $y=2x \Rightarrow x=\frac{y}{2}$

The equation is $\left(\frac{y}{2}\right)^4 + 4\left(\frac{y}{2}\right)^3 - 3\left(\frac{y}{2}\right)^2 - 4\left(\frac{y}{2}\right) + 2 = 0$

$\therefore y^4 + 8y^3 - 12y^2 - 32y + 32 = 0$ ✗

(ii) let $y=\frac{1}{\sqrt{x}} \Rightarrow x=\frac{1}{\sqrt{y}}$

The equation is $\left(\frac{1}{\sqrt{y}}\right)^4 + 4\left(\frac{1}{\sqrt{y}}\right)^3 - 3\left(\frac{1}{\sqrt{y}}\right)^2 - 4\left(\frac{1}{\sqrt{y}}\right) + 2 = 0$

$\frac{1}{y^2} + \frac{4}{y\sqrt{y}} - \frac{3}{y} - \frac{4}{\sqrt{y}} + 2 = 0$

$1 + 4\sqrt{y} - 3y - 4y\sqrt{y} + 2y^2 = 0$

$(1 - 3y + 2y^2)^2 = 4\sqrt{y}(y-1)$

$\therefore (1 - 3y + 2y^2)^2 = 16y(y-1)^2$

$\therefore 1 - 6y + 13y^2 - 12y^3 + 4y^4 = 16y^3 - 32y^2 + 16y$

$\therefore 4y^4 - 28y^3 + 45y^2 - 22y + 1 = 0$ ✗

(e) (i) let $x = \cos \theta$

$\cos 5\theta = 16x^5 - 20x^3 + 5x$

$\therefore 16x^5 - 20x^3 + 5x - 1 = 0$

$\Rightarrow 16x^5 - 20x^3 + 5x = 1$

$\Rightarrow \cos 5\theta = 1$

$5\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi$

$\theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$

$\therefore x = \cos \theta, \cos \frac{2\pi}{5}, \cos \frac{4\pi}{5}, \cos \frac{6\pi}{5}, \cos \frac{8\pi}{5}$
 $= 1, \cos \frac{2\pi}{5}, \cos \frac{4\pi}{5}, \cos \frac{4\pi}{5}, \cos \frac{2\pi}{5}$

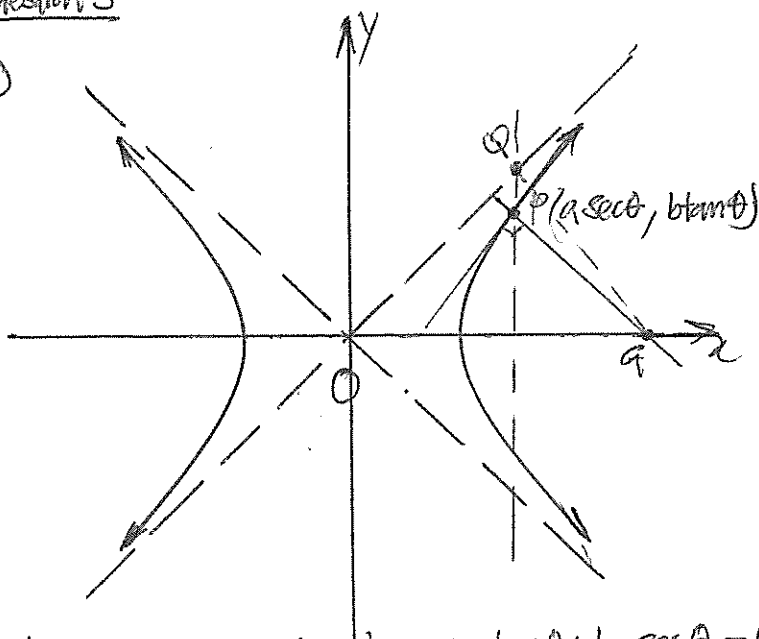
(ii) $1 + \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} + \cos \frac{4\pi}{5} + \cos \frac{2\pi}{5} = 0$

$2(\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5}) = -1$

$\therefore \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$ ✗

Question 3

(a)



Normal to hyperbola is $ax \tan \theta + by \sec \theta = (a^2 + b^2) \sec \theta \tan \theta$

When $y=0$, $ax \tan \theta = (a^2 + b^2) \sec \theta \tan \theta$

$$ax = (a^2 + b^2) \sec \theta$$

$$x = \frac{(a^2 + b^2) \sec \theta}{a}$$

$$\therefore Q \left(\frac{(a^2 + b^2) \sec \theta}{a}, 0 \right)$$

Asymptote has equation $y = \frac{b}{a} x$

When $x = a \sec \theta$, $y = \frac{b}{a} \cdot a \sec \theta = b \sec \theta$

$\therefore Q$ is $(a \sec \theta, b \sec \theta)$

$$m_{OQ} = \frac{b}{a}$$

$$m_{AQ} = \frac{b \sec \theta - 0}{a \sec \theta - \frac{(a^2 + b^2) \sec \theta}{a}}$$

$$= \frac{ab \sec \theta}{a^2 \sec \theta - a^2 \sec^2 \theta - b^2 \sec \theta}$$

$$= \frac{ab \sec \theta}{a^2 \sec \theta - a^2 \sec^2 \theta - b^2 \sec \theta}$$

$$= \frac{ab \sec \theta}{-b^2 \sec \theta}$$

$$= -\frac{a}{b}$$

$\therefore m_{OQ} \times m_{AQ} = \frac{b}{a} \times -\frac{a}{b} = -1 \Rightarrow OQ$ is at right angles to the asymptote.

$$(i) y = \frac{c}{x} \Rightarrow y' = -\frac{c^2}{x^2}$$

$$m_{\text{tangent}} = \frac{-c^2}{(ct)^2} = -\frac{1}{t^2}$$

$$\text{Equation of tangent is } y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$$

$$\text{i.e. } x + t^2 y = 2ct$$

Gradient of ON is t^2 & has equation $y = t^2 x$

$$N: x + t^2(t^2 x) = 2ct$$

$$x(1+t^4) = 2ct$$

$$x = \frac{2ct}{1+t^4} \quad \& \quad y = \frac{2ct^3}{1+t^4}$$

$$\therefore N \text{ is } \left(\frac{2ct}{1+t^4}, \frac{2ct^3}{1+t^4} \right)$$

(ii) Show that the point lies on $(x^2 + y^2)^2 = 4c^2 xy$

$$\text{LHS} = (x^2 + y^2)^2$$

$$= \left(\left(\frac{2ct}{1+t^4} \right)^2 + \left(\frac{2ct^3}{1+t^4} \right)^2 \right)^2$$

$$= \left(\frac{4c^2 t^2 + 4c^2 t^6}{(1+t^4)^2} \right)^2$$

$$= \left[\frac{4c^2 t^2 (1+t^4)}{(1+t^4)^2} \right]^2$$

$$= \left[\frac{4c^2 t^2}{(1+t^4)} \right]^2$$

$$= \frac{16c^4 t^4}{(1+t^4)^2}$$

$$\text{RHS} = 4c^2 xy$$

$$= 4c^2 \cdot \frac{2ct}{(1+t^4)} \cdot \frac{2ct^3}{(1+t^4)}$$

$$= \frac{16c^4 t^4}{(1+t^4)^2}$$

$$= \text{LHS}$$

$\therefore N$ lies on the curve

$$(x^2 + y^2)^2 = 4c^2 xy \quad \checkmark$$

$$(d) \quad xy(x+y)+16=0$$

$$x^2y+xy^2+16=0$$

$$x^2 \cdot \frac{dy}{dx} + y \cdot 2x + x \cdot 2y \frac{dy}{dx} + y^2 \cdot 1 = 0$$

$$\frac{dy}{dx}(x^2+2xy) + 2xy+y^2 = 0$$

$$\frac{dy}{dx} = - \frac{2xy+y^2}{x^2+2xy}$$

$$\text{At } (-2, -2)$$

$$\frac{dy}{dx} = m_{\text{tangent}} = - \frac{2 \cdot (-2) \cdot (-2) + (-2)^2}{(-2)^2 + 2 \cdot (-2) \cdot (-2)} = - \frac{12}{12} = -1$$

$$\text{Equation of tangent is } y+2 = -1(x+2)$$

$$y = -x-4$$

Question 4

(a) & (b) — check answers using GeoGebra.

Question 5

$$(a) (i) \text{ Let } \angle SPT = \alpha^\circ$$

$$\angle PRS = \alpha^\circ \text{ (alternate segment theorem)}$$

$$PT = QT \text{ (intercepts on tangents from an external point are equal)}$$

$$\therefore \angle PQT = \alpha^\circ \text{ (angles opposite equal sides are equal)}$$

$$\therefore PT \text{ subtends equal angles at } R \text{ \& } Q \text{ on the same side of it}$$

$$\therefore P, R, Q, T \text{ is a cyclic quadrilateral.}$$

$$(ii) \angle QRT = \angle QPT \text{ (angles at circumference standing on arc } QT \text{ are equal)}$$

$$= \alpha^\circ$$

$$\therefore \angle QRS = \angle PRS$$

$$\therefore RS \text{ bisects } \angle QRP.$$

$$(b) (i) \sqrt{n+1} - \sqrt{n} = \sqrt{n+1} - \sqrt{n} \times \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

$$\begin{aligned}
 \text{Consider } \sqrt{n+1} - \sqrt{n} - \frac{1}{2\sqrt{n+1}} &= \frac{1}{\sqrt{n+1} + \sqrt{n}} - \frac{1}{2\sqrt{n+1}} \\
 &= \frac{2\sqrt{n+1} - (\sqrt{n+1} + \sqrt{n})}{2\sqrt{n+1} [\sqrt{n+1} + \sqrt{n}]} \\
 &= \frac{\sqrt{n+1} - \sqrt{n}}{2\sqrt{n+1} [\sqrt{n+1} + \sqrt{n}]}
 \end{aligned}$$

$$\sqrt{n+1} > \sqrt{n} \text{ for } n > 0$$

$$\sqrt{n+1} > 0, \sqrt{n} > 0$$

$$\therefore \sqrt{n+1} + \sqrt{n} > 0$$

$$> 0$$

$$\therefore \sqrt{n+1} - \sqrt{n} > \frac{1}{2\sqrt{n+1}}$$

(II) Prove true for $n=7$

$$\text{LHS} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} = \frac{363}{140} \approx 2.592857143$$

$$\text{RHS} = \sqrt{7} \approx 2.645751311$$

$$\therefore 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} < \sqrt{7} \quad \therefore \text{True for } n=7$$

Assume true for $n=k$

$$\text{i.e. } 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{k} < \sqrt{k}$$

Prove true for $n=k+1$

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{k} + \frac{1}{k+1} < \sqrt{k} + \frac{1}{k+1}$$

Consider $\sqrt{k+1} - \sqrt{k}$

$$\text{We know } \sqrt{k+1} - \sqrt{k} > \frac{1}{2\sqrt{k+1}}$$

$$\text{Consider } \frac{1}{2\sqrt{k+1}} - \frac{1}{k+1} = \frac{k+1 - 2\sqrt{k+1}}{2\sqrt{k+1}(k+1)} = \frac{\sqrt{k+1} [\sqrt{k+1} - 2]}{2(k+1)(\sqrt{k+1})} > 0$$

$$\text{Since } k \geq 7, \sqrt{k+1} - 2 > 0$$

$$\therefore \frac{1}{2\sqrt{k+1}} > \frac{1}{k+1}$$

$$\& \sqrt{k+1} > 0, (k+1) > 0$$

Back to the question ...

$$\therefore \sqrt{k} + \frac{1}{|k+1|} < \sqrt{k+1}$$

$$\therefore 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{k+1} < \sqrt{k+1}$$

$\therefore$ True for $n = k+1$

$\therefore$ True by Mathematical Induction for $n \geq 7$.

(c) (i) $y = e^x - x - 1$

$$y' = e^x - 1$$

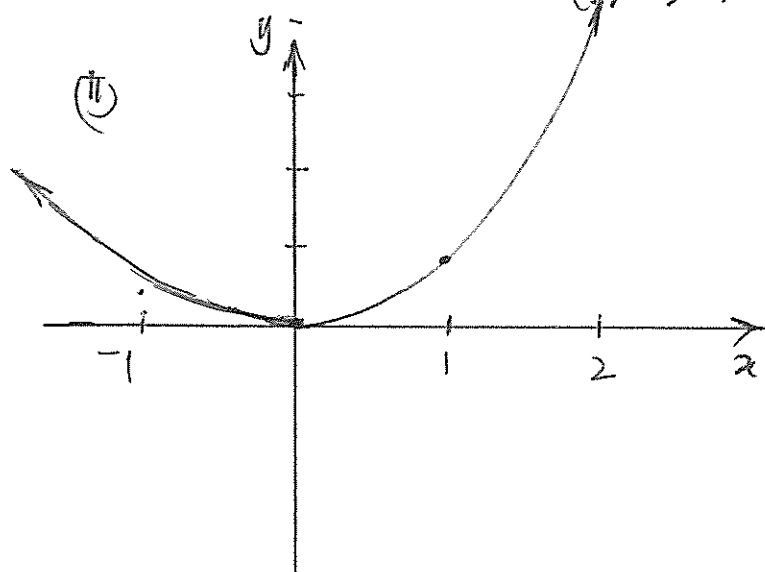
$$y'' = e^x$$

Stationary points $y' = 0 \Rightarrow e^x - 1 = 0$

$$e^x = 1$$

$$x = 0$$

$\therefore$ At $(0, 0)$, $y'' = e^0 = 1 > 0 \therefore$ Relative minimum



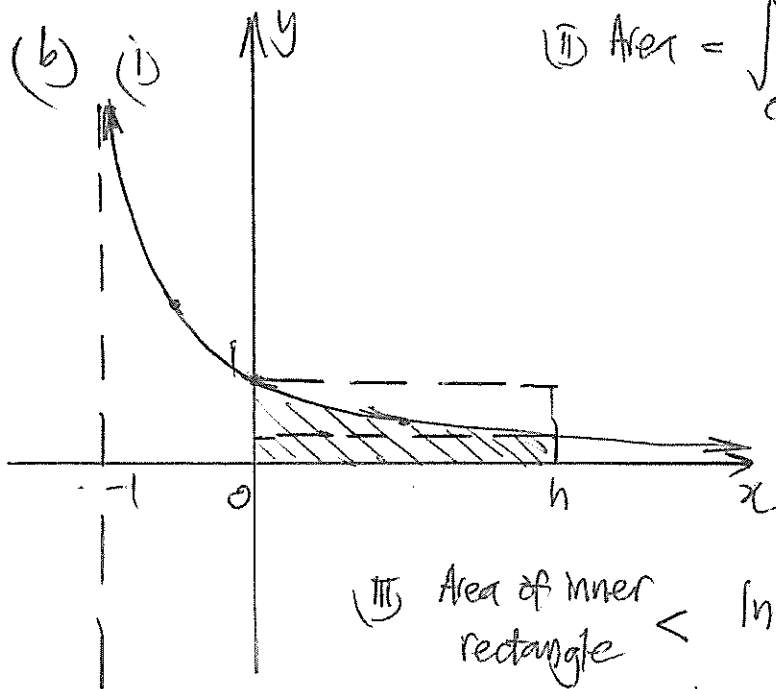
(iii) From the graph

$$e^x - x - 1 \geq 0 \text{ for all } x$$

$$\therefore e^x \geq x + 1.$$

Question 6

$$\begin{aligned} (a) \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h} \\ &= e^x \lim_{h \rightarrow 0} \frac{1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \dots - 1}{h} \\ &= e^x \lim_{h \rightarrow 0} \frac{h + \frac{h^2}{2!} + \frac{h^3}{3!} + \dots}{h} \\ &= e^x \lim_{h \rightarrow 0} \left(1 + \frac{h}{2!} + \frac{h^2}{3!} + \dots \right) \\ &= e^x \cdot 1 = e^x \end{aligned}$$



$$\begin{aligned} \text{(ii) Area} &= \int_0^h \frac{1}{1+x} dx = \left[\ln(1+x) \right]_0^h \\ &= \ln(1+h) - \ln(1) \\ &= \ln(1+h) \end{aligned}$$

(iii) Area of inner rectangle $< \ln(1+h) < \text{Area of outer rectangle}$

$$h \times \frac{1}{1+h} < \ln(1+h) < 1 \cdot h$$

$$\therefore \frac{h}{1+h} < \ln(1+h) < h$$

(iv) Divide the result of (iii) by h

$$\frac{1}{1+h} < \frac{\log_e(1+h)}{h} < 1$$

As $h \rightarrow 0$ $\frac{1}{1+h} \rightarrow 1$ $\therefore \lim_{h \rightarrow 0} \frac{\log_e(1+h)}{h} = 1$

$\left(\frac{\log_e(1+h)}{h} \right)$ is sandwiched between two values both approaching 1

(v) $\lim_{h \rightarrow 0} \frac{\log_e(1+h)}{h} = 1$

Let $h = \frac{1}{x}$ $\therefore \lim_{x \rightarrow \infty} \frac{\log_e\left(1+\frac{1}{x}\right)}{\frac{1}{x}} = 1$

As $h \rightarrow 0$, $x \rightarrow \infty$

$$\therefore \lim_{x \rightarrow \infty} x \log_e\left(1+\frac{1}{x}\right) = 1$$

(c) (i) Consider $\frac{a+b}{2} - \sqrt{ab}$

$$= \frac{a+b-2\sqrt{ab}}{2}$$

$$= \frac{(\sqrt{a}-\sqrt{b})^2}{2}$$

$$\geq 0$$

$$\therefore \frac{a+b}{2} \geq \sqrt{ab}$$

$$(ii) \frac{a+b+c+d}{4} = \frac{\frac{a+b}{2} + \frac{c+d}{2}}{2} \geq \sqrt{\left(\frac{a+b}{2}\right)\left(\frac{c+d}{2}\right)}$$

$$\frac{a+b}{2} \geq \sqrt{ab}$$

$$\geq \sqrt{\sqrt{ab} \cdot \sqrt{cd}}$$

$$\frac{c+d}{2} \geq \sqrt{cd}$$

$$= \sqrt{\sqrt{abcd}}$$

$$= \sqrt[4]{abcd}$$